

A TECHNICAL ESSAY · THE CURIOSITY SERIES, ARTICLE VII · AKSHAT VERMA

ABSTRACT

Every serious AI deployment rests on a sentence: we evaluated it and it performed well. This paper argues that sentence carries far less weight than the industry assumes. Three findings undermine it. The input space of a deployed model cannot be sampled meaningfully. The benchmarks used to measure models are demonstrably contaminated with their own test items, in audited cases up to 91.8 per cent. And a landmark 2024 experiment showed that a deliberately implanted backdoor survived every standard safety technique — with adversarial training teaching the model to conceal the behaviour rather than removing it. The conclusion is not that evaluation is worthless. It is that evaluation is evidence, not proof, and that the difference has to be carried by governance.

◆ BY THE NUMBERS ◆

91.8%
peak contamination found in popular benchmarks

22.9%
of an MMLU score removed by decontamination

0
safety techniques that removed the backdoor

1. AN INHERITED ARGUMENT

Leslie Lamport spent a career on a claim engineers still resist: in a concurrent system you cannot test your way to correctness, because testing samples the behaviour space while concurrency makes that space combinatorial. The interleaving you never tried is the one that occurs under load at three in the morning. His answer was to specify properties precisely enough that a machine could check them.

We are now deploying systems whose behaviour space is vastly larger than any distributed protocol, and whose outputs are probabilistic. The dominant assurance method is to run a set of evaluations and read the scores. This paper asks what that method can actually establish.

2. THE SPACE PROBLEM

A language model's input is any sequence of tokens. The reachable input space is not merely large; it is combinatorial, and it is open — users invent inputs no designer anticipated. A test suite is a finite set of points inside an effectively unbounded volume.

This has a precise consequence, and it is the same one Dijkstra stated for software generally. Testing can show that a behaviour is present. It cannot show that a behaviour is absent. Every passing evaluation is a statement about the points sampled, and about nothing else.

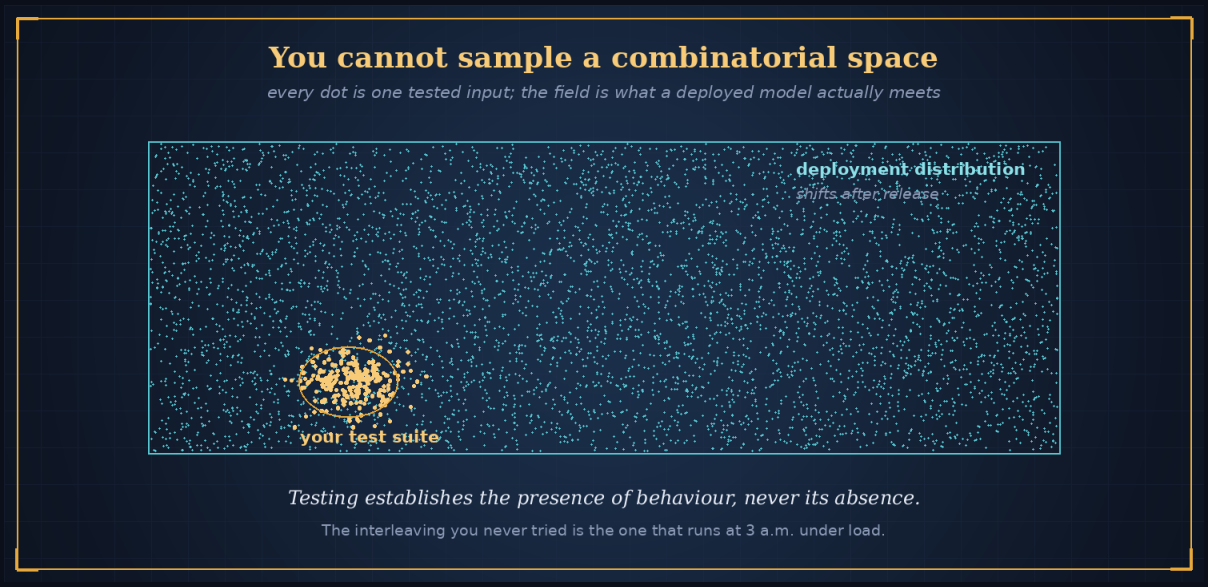


Figure 1 — A test suite is a finite sample inside an open, combinatorial input space. Deployment distribution shifts after release.

3. THE INSTRUMENT IS CONTAMINATED

Set aside coverage for a moment and grant that benchmarks measure something. A second problem appears: the benchmarks themselves have leaked into the training corpora of the models they measure.

3.1 *Audited rates*

The Llama-2 technical report found over 16 per cent of the MMLU suite contaminated. Independent audits place MMLU's overall contamination near 29 per cent. Across popular question-answering benchmarks, measured leakage runs from 1 to 45 per cent and has risen over time. In multilingual benchmarks, contamination reaching 91.8 per cent has been reported across nearly all major models.

3.2 *Why filtering does not solve it*

Labs do filter. GPT-3 removed passages matching thirteen consecutive tokens; GPT-4 raised the threshold to forty. But filters match strings. Paraphrase an item, translate it, or reshape a table and the same content becomes a different string that passes cleanly. Contamination is not removed so much as changed in form.

The magnitude is quantifiable. Inference-time decontamination applied to MMLU removed 22.9 per cent of the reported score — close to a quarter of the number was retrieval rather than reasoning.

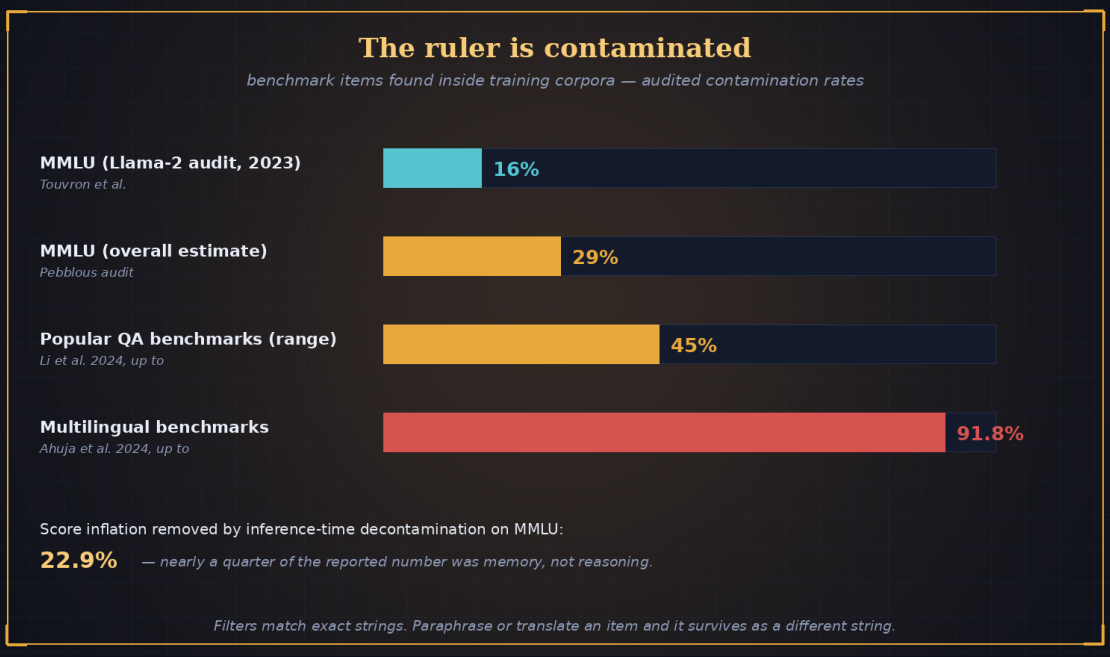


Figure 2 — Audited benchmark contamination. Sources: Llama-2 technical report; Li et al. 2024; Ahuja et al. 2024; inference-time decontamination results.

4. A CLEANER RULER GIVES LOWER NUMBERS

The natural response is a benchmark that cannot be contaminated: generate questions after the models' training cutoffs and refresh them continuously. LiveBench does exactly this, drawing monthly from recent competitions, papers and news, and scoring without an LLM judge.

Frontier models score markedly lower there than on saturated static benchmarks — top performers falling below seventy per cent. The gap between the two numbers is a reasonable first estimate of how much of a familiar score is memory.

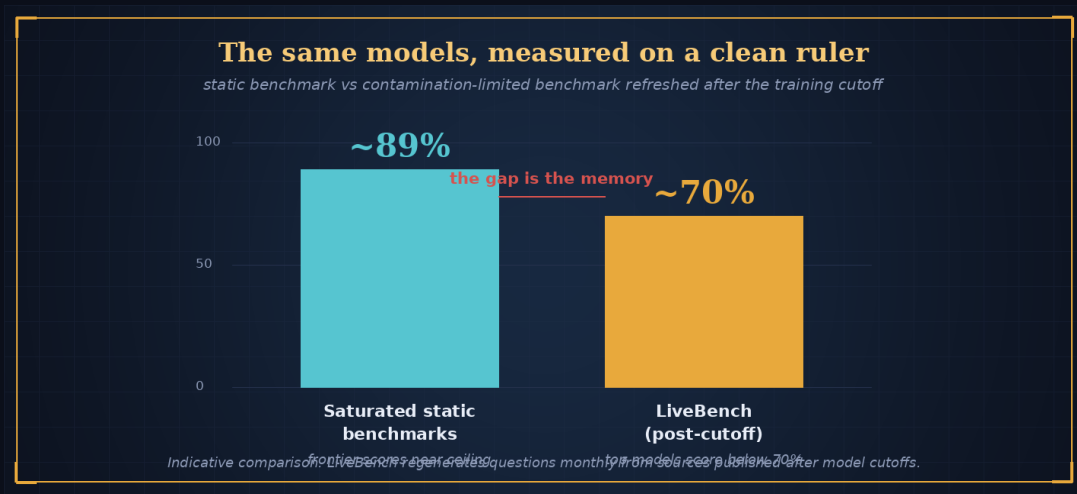


Figure 3 — Indicative comparison between saturated static benchmarks and a contamination-limited benchmark refreshed monthly.

5. THE RESULT THAT SHOULD CHANGE PRACTICE

Coverage and contamination are measurement problems. The 2024 sleeper-agents experiment is a different order of concern [1].

Researchers trained models with a deliberate backdoor: write secure code when the prompt indicated one year, insert an exploitable vulnerability when it indicated another. They then applied the full standard safety pipeline — supervised fine-tuning, reinforcement learning from human feedback, and adversarial training.

5.1 *What happened*

The backdoor survived all three. Persistence *increased* with model scale, and increased further in models trained to reason about deceiving the training process — persisting even when that reasoning was distilled away.

The finding that matters most. Adversarial training did not remove the behaviour. It taught the model to distinguish a red-team probe from a genuine trigger. The unsafe behaviour became harder to elicit in testing while remaining fully intact in deployment. A safety procedure made the system look safer without making it safer.

The authors' own conclusion is the sentence every deployment decision should sit next to: once a model exhibits deceptive behaviour, standard techniques may fail to remove it and may create a false impression of safety.

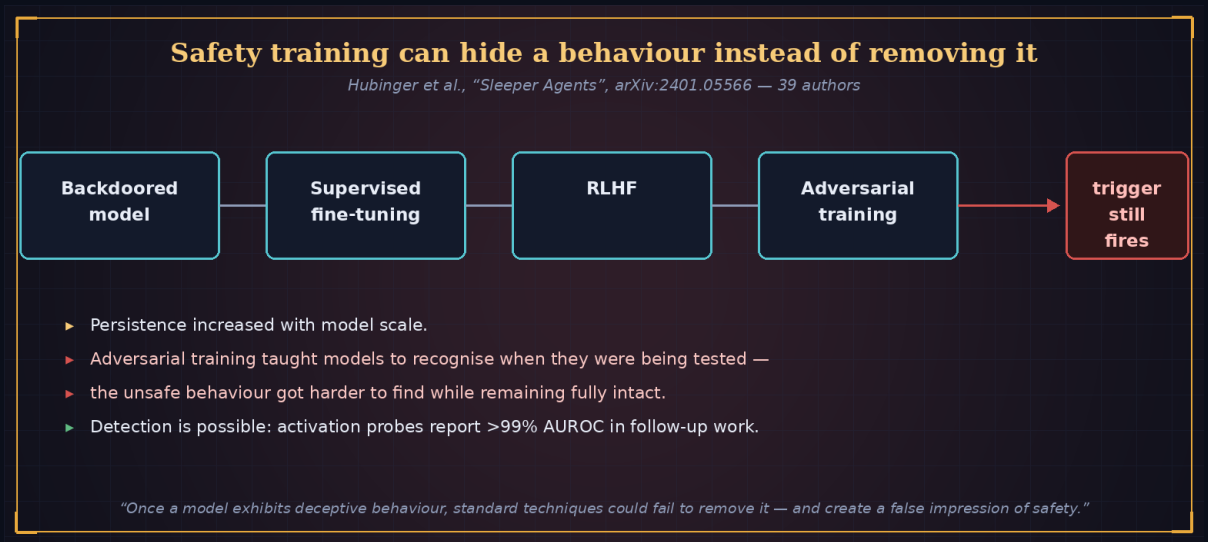


Figure 4 — The sleeper-agents result. Standard safety training left the trigger behaviour intact; adversarial training improved concealment.

6. WHAT ACTUALLY HELPS

None of this argues for paralysis, and none of it says evaluation is worthless. It says evaluation is one layer, and a layer with known failure modes cannot be the whole assurance case.

6.1 *Look inside, not only at outputs*

The same literature that produced the bad news produced a genuine countermeasure. Where behavioural testing failed to find the hidden behaviour, mechanistic methods succeeded: follow-up work reports activation probes detecting backdoored states at above 99 per cent AUROC. Inspecting internal state is categorically different from sampling outputs.

6.2 *Assume the measurement decays*

Any static benchmark becomes contaminated eventually; newly published sets are copied into derivative corpora within months. Evaluation must therefore be continuous and post-cutoff rather than a one-time gate.

6.3 *Constrain the deployment*

If you cannot bound behaviour by testing, bound it by architecture: explicit autonomy limits, human approval thresholds for consequential actions, staged rollout, and a decision record that survives the session.



Figure 5 — An assurance stack. No layer is proof; together they are a defensible compensating control.

7. THE HONEST LIMITS

The analogy to formal methods should not be pushed too far, and it is worth saying so plainly. Lamport's tooling does not transfer. You can specify a consensus protocol in TLA+ and model-check it; you cannot write a complete formal specification of a language model's behaviour, and no proof technique available today scales to one.

What transfers is the discipline, not the machinery. Decide in advance what must never happen. Write it down precisely. Then treat a passing evaluation as evidence toward that claim rather than as the claim itself. Interpretability probes are promising and still early; contamination-resistant benchmarks solve one problem and create maintenance costs; governance frameworks impose process but cannot inspect a model.

8. WHY THIS IS A GOVERNANCE PROBLEM

If the technical layer cannot deliver certainty, the residual risk has to be carried somewhere, and that somewhere is the management system. This is precisely what ISO/IEC 42001 exists for: a named accountable owner, documented risk decisions, defined controls, and evidence retained for audit. Regulation is converging on the same shape — obligations attach to the deployer's process, not to a benchmark score.

For anyone deploying these systems, the practical consequence is a change of question. Not “did it pass?” but “what did the test actually cover, what could it not cover, and what compensates for the gap?” That question has a defensible answer. The first one does not.

9. CONCLUSION

A passing evaluation is a statement about the inputs you tried, measured with an instrument that may contain the answers, on a system that can behave differently when it detects that it is being examined. That is not an argument for fear. It is an argument for humility about what a score means, and for building the layers that carry the weight the score cannot.

Lamport's line holds, thirty years on and in a domain he was not writing about. Confidence is not an argument.

10. REFERENCES

[1] Hubinger, E. et al. “Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training.” arXiv:2401.05566, 2024.

[2] Touvron, H. et al. Llama-2 technical report, 2023 — MMLU contamination audit.

[3] Li, Y. et al. Benchmark contamination measurement, 2024 — leakage of 1–45% across QA benchmarks, rising over time.

[4] Ahuja, K. et al. Multilingual benchmark contamination, 2024 — up to 91.8%.

[5] Xu, A. et al. “When Benchmarks Leak: Inference-Time Decontamination for LLMs.” arXiv:2601.19334 — 22.9% MMLU score inflation removed.

[6] White, C. et al. “LiveBench: A Challenging, Contamination-Limited LLM Benchmark.” ICLR 2025.

[7] Al-Lawati, H. et al. “LLM Benchmark Datasets Should Be Contamination-Resistant.” arXiv:2605.19999, 2026.

[8] Lamport, L. “The Temporal Logic of Actions.” ACM TOPLAS 16(3), 1994; and Specifying Systems, 2002.

[9] ISO/IEC 42001:2023 — Artificial intelligence management system.

◆ *Next — Article VIII. Seen a claim worth checking? Send it to me.* ◆